

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Lesson 7: Trees and Ensembles

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

20 November 2026

Agenda

- A fourth family, built on a threshold
- Decision trees, and why one alone is mediocre
- Depth as the bias-variance dial, again
- Bagging and random forests
- Feature importance, and its limits
- Gradient boosting
- Choosing among trees, forests and boosting

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Exercise 6, before we start

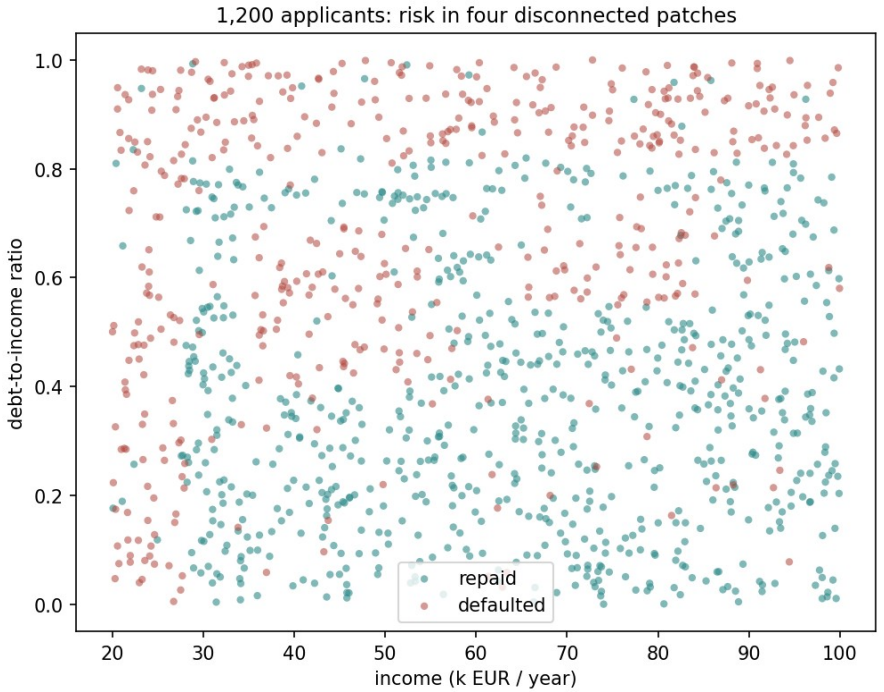
- What counts is the **defence of which family**, not the winning score
- The gap to watch for: a strong accuracy number with no check of *why* the assumption behind it held
- Today’s models come with the same obligation

1,200 loan applicants, two figures each

- Income in thousands of euros a year, and debt-to-income ratio
- Underwriting is a **checklist**: an income floor, a debt ceiling, and two “stressed” combinations, each figure ordinary alone but risky together
- **7%** of labels flipped: no honest model beats about **0.93**, the **noise ceiling**
- **38.7%** defaulted, so the majority baseline is **0.613**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Four disconnected risky regions



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

No straight line separates them

Model	Cross-validated accuracy
Always predict the majority class	0.613
Logistic regression	**0.748 ± 0.030**

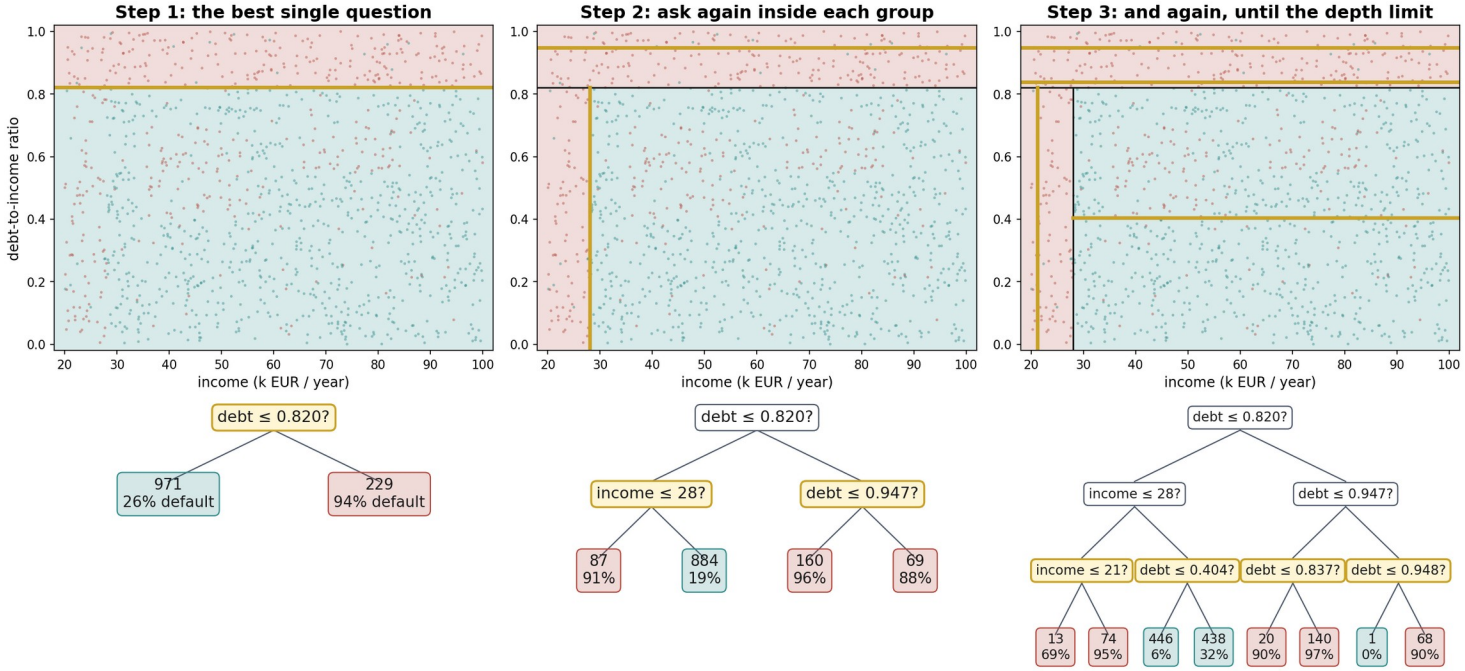
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The whole algorithm, in one sentence

- Find the (feature, threshold) split that makes the two groups **as pure as possible**
- Apply it; repeat on each group, independently
- Stop at a depth limit, a pure group, or too few examples to split
- No coefficients and no gradient: each threshold is chosen by **searching** the data
- **Greedy**: whichever split helps most *right now*, with no lookahead

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The algorithm, drawn



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How pure is a group of examples?

- A leaf that is all one class needs no more questions
- A leaf split evenly between classes is the worst case: a coin flip
- **Impurity** is a number that captures this: 0 for a pure group, highest for an even mix. “As pure as possible” then means “with the lowest impurity”

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Gini impurity

$$G = 1 - \sum_{c=1}^{C} p_c^2$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

For two classes, one number

- G = 2p(1-p), where p is the fraction of one class
- **0** when the group is pure (p = 0 or p = 1)
- **0.5**, the maximum, at p = 0.5
- On the full 1,200 applicants, p = 0.387 gives G = 0.474

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

The gain a split buys

- A candidate split divides a parent group P into a left child L and a right child R; *m* counts the examples in each
- Worth taking if the children are, **on average, purer** than the parent
- The average is **weighted by size**: one example cut off alone would halve a plain average, and barely moves a weighted one

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The reduction in impurity

$$\Delta G = G_P - \left(\frac{m_L}{m_P} G_L + \frac{m_R}{m_P} G_R \right)$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Scanning for the best split

- For a continuous feature, scan **every midpoint** between adjacent sorted values as a candidate threshold
- Do this for **every feature**; keep the best pair
- Repeat the search inside each child
- **Greedy**: ΔG is judged one split at a time, never jointly

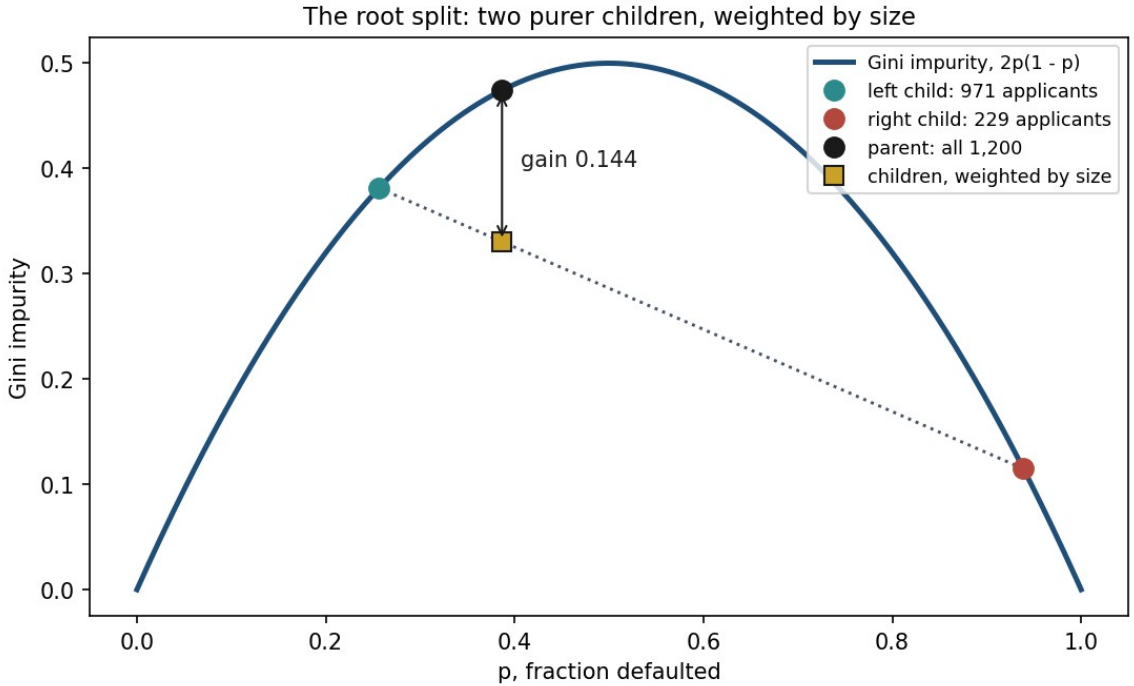
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Worked example: the root split

- p = 0.387 defaulted, so the parent group’s impurity is **0.474**
- Exhaustive search finds `debt_ratio <= 0.82`: the **debt ceiling** the data was built with
- 971 applicants left (0.256 defaulted, impurity 0.381), 229 right (0.939, impurity 0.115)
- Weighted by size, the children average 0.3305 against the parent’s 0.4743: gain **0.144**

UNIVERSITÀ DEGLI STUDI DELL’AQUILA

The root split, on the Gini curve



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Checked against scikit-learn

- The from-scratch implementation and scikit-learn's `DecisionTreeClassifier` make **the same prediction for all 1,200 applicants**, at every depth tested
- Not a coincidence: both search the same space by the same greedy rule
- From here on, the notebooks use scikit-learn's implementation: the same algorithm, compiled, not a different one

What greedy misses

- The search never asks “which pair of splits helps most together”: only “which single split helps most right now”
- A feature that only pays off **in combination** with a later split can be invisible to a shallow tree
- Exactly what happens to this dataset’s two interior “stressed” islands, covered next

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Depth is the bias-variance dial

- `max_depth`: the most questions **one applicant** answers
- **Shallow tree**: few questions, only the coarsest structure. High bias, low variance
- **Deep tree**: enough questions to isolate single points. Training accuracy **exactly 1.000**, even on data with no structure, unless two examples share every feature value but not the label

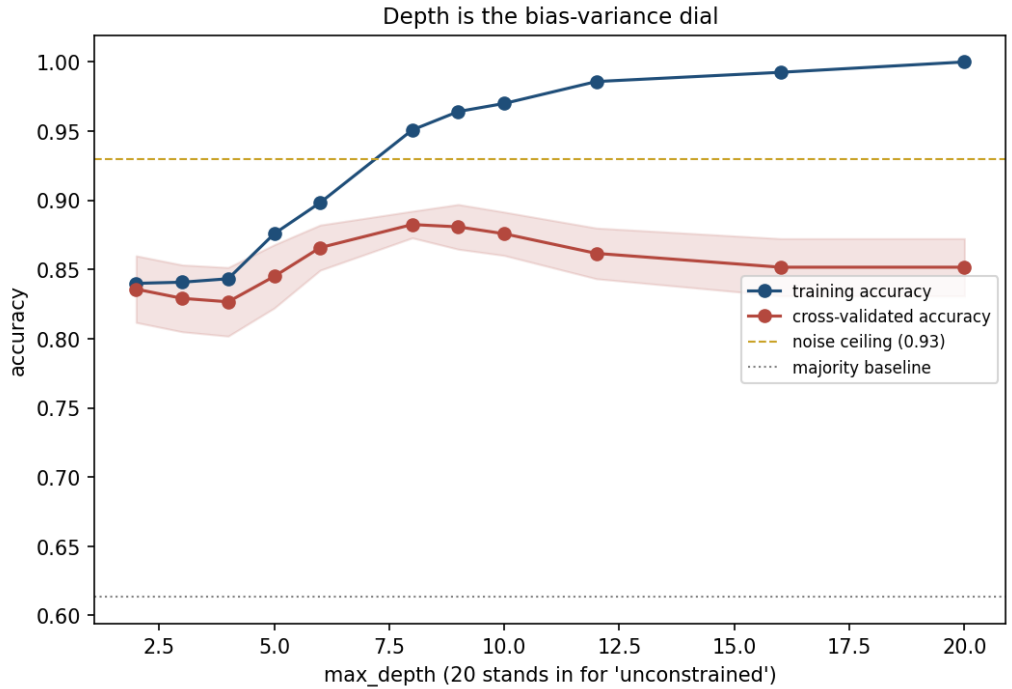
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Measured, at seven depths

max_depth	Training accuracy	Cross-validated accuracy
2	0.840	0.836 ± 0.024
4	0.843	0.827 ± 0.025
6	0.898	0.866 ± 0.016
8	0.951	**0.882 ± 0.010**
12	0.986	0.862 ± 0.018
16	0.993	0.852 ± 0.021
unconstrained	**1.000**	0.852 ± 0.021

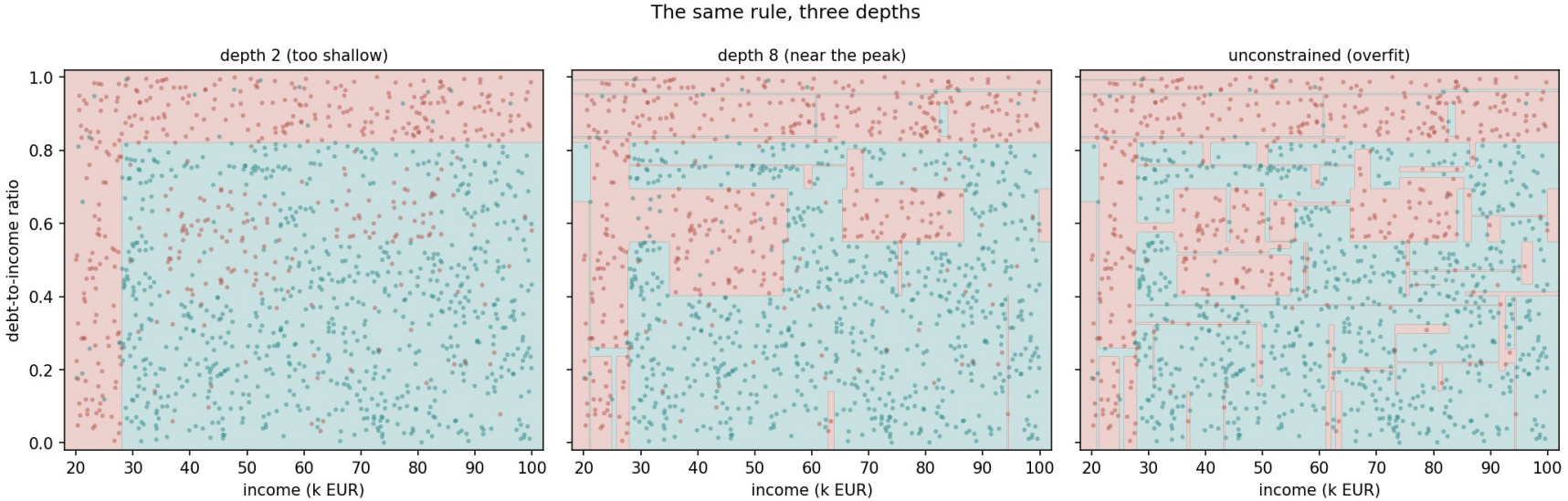
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The curve bends downward



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The same rule, three depths



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Reading a tree, and the knob that keeps it small

- `max_depth` **caps every path**: each node above the cap is split if it can be, worth it or not
- `max_leaf_nodes` is a **budget of leaves**, spent on the largest ΔG first
- `max_leaf_nodes = 9`: **90.8%** training accuracy; thresholds 28 (income floor) and 0.82 (debt ceiling)
- In code: `export_text(tree)` prints the tree as if/else rules

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

A tree that doesn’t stay still

- Two trees at `max_depth = 6`, each on a random 65% of the applicants (no replacement)
- The debt ceiling, the strongest split, agrees to within **0.001**
- The trees grow **31 and 33 leaves**, and **disagree on 11.7%** of predictions on the full dataset
- The strongest split is stable; the weaker ones, which shape most of the tree, are not

UNIVERSITÀ DEGLI STUDI DELL’AQUILA

Notebook 1, live

section, and its slides	you call (→ the number to come back with)
1. look at it first, 4-6	`make_pipeline(StandardScaler(), LogisticRegression())` in `cross_val_score(..., cv=StratifiedKFold(5))` → 0.613 against 0.748
2. a tree, split by split, 7-18	your own `gini` and `best_split`, then `DecisionTreeClassifier(max_depth=d)` → 100.0% identical predictions; gain 0.1438
3. the depth dial, 19-22	`DecisionTreeClassifier(max_depth=d)` in `cross_val_score` → peak 0.882 at depth 8
4. what a tree costs, 23-24	`DecisionTreeClassifier(max_leaf_nodes=9)`, `export_text` → 0.908; two trees agree on 0.883

Break

- Twelve minutes

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Bagging: averaging away the instability

- Trees differ across samples, so **many trees, averaged** should partly cancel their mistakes
- **Bootstrap sample**: m rows drawn **with replacement** from the m training rows
- One tree per sample, `n_estimators` times; predict by **majority vote**. That is **bagging**: bootstrap aggregating
- In code: `BaggingClassifier()`, whose default model is a decision tree

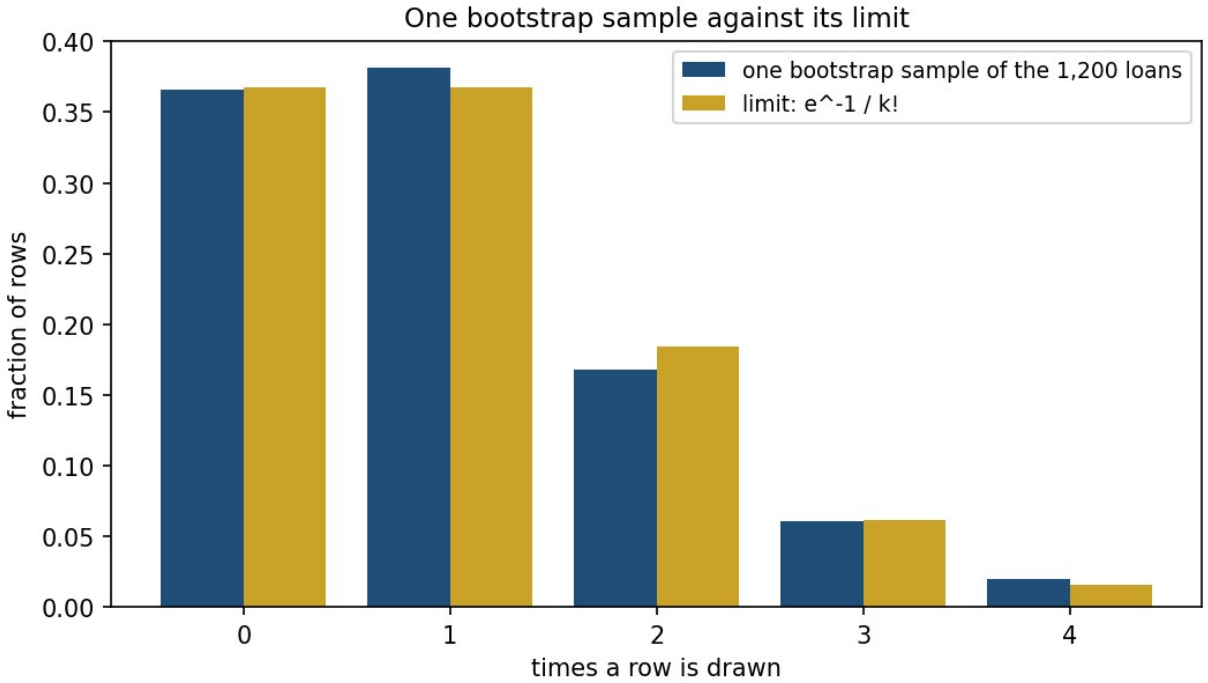
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How much data one bootstrap sample sees

- One row is missed by all m draws with probability $(1 - 1/m)^m$, which tends to $e^{-1} \approx 0.368$ as m grows
- So about **63.2%** of the distinct rows are drawn at least once
- The remaining **36.8%** are left out entirely: called **out-of-bag (OOB)**
- At m = 1,200: a single simulated draw left out 36.6% of rows, close to the limit

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

One bootstrap sample, counted



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Out-of-bag: free validation

- Predict each row by a vote of only the trees that never saw it: the accuracy of those votes is the **OOB score**, with no rows set aside
- A 300-tree random forest (bagged trees plus one rule, slide 34): OOB **0.9117**; 5-fold cross-validation **0.9117 ± 0.021**
- The four-decimal match is **luck**: over twelve seeds the two differ by 0.0026 on average, well inside ±0.021

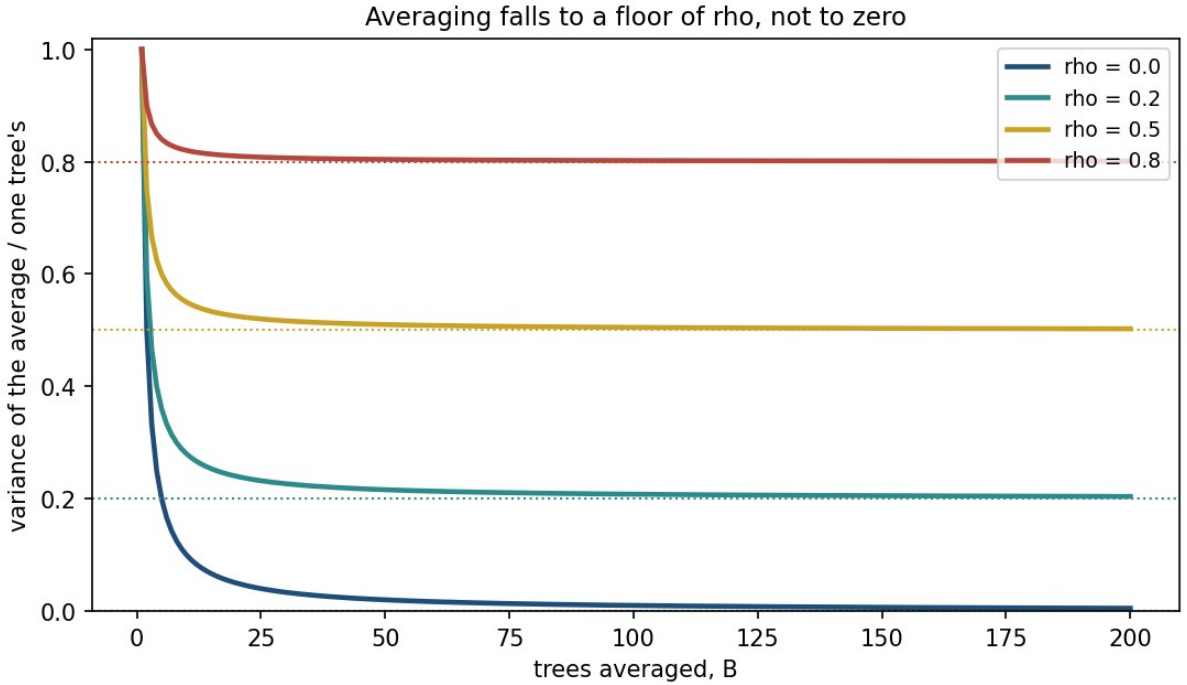
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Why averaging reduces variance

- Model each tree’s prediction as noisy, with variance σ^2 over the bootstrap sample it drew, and let ρ be the correlation between two trees
- The average of B trees has variance $\sigma^2/B + \rho\sigma^2 (B-1)/B$: a **floor** of $\rho\sigma^2$, **not zero**: more trees soon buy little
- Averaging **cannot repair an error every tree shares**: trees too shallow for the rule stay too shallow, however many are averaged

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Averaging stops at a floor



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Bagging, measured

Model	Mean test accuracy, 30 random 70/30 splits	Standard deviation across the 30
Single unconstrained tree	0.856	0.0185
Bagging, 100 trees	**0.902**	**0.0153**

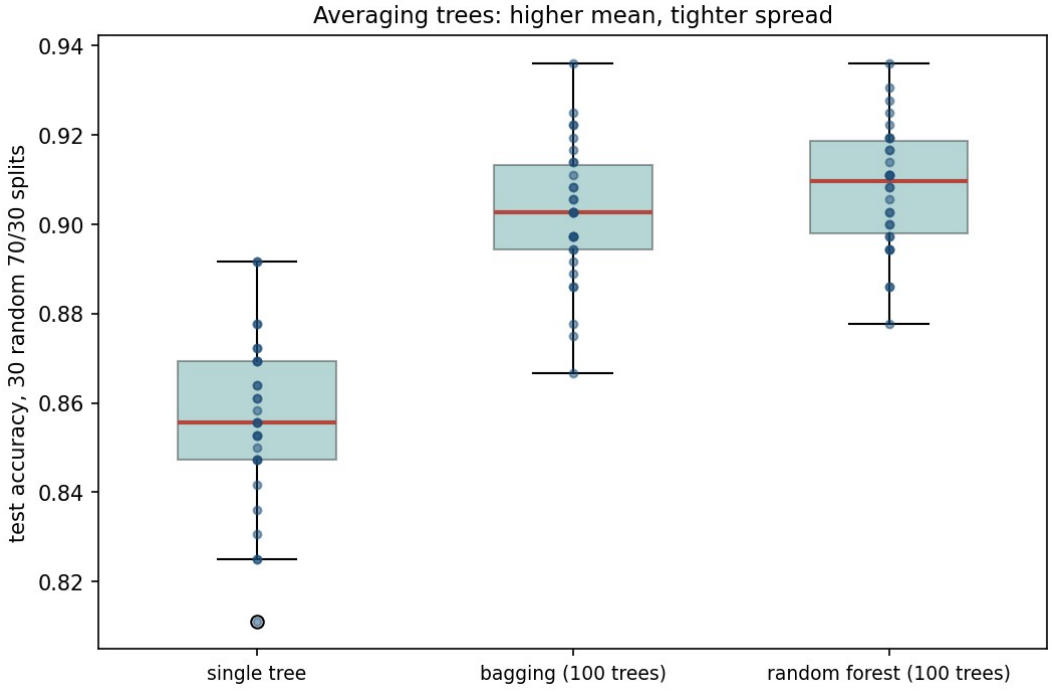
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Random forests: decorrelating the trees

- The floor is $\rho\sigma^2$: lowering it means making the trees agree **less**
- Bagged trees see **every feature**, so a strong one pulls them all the same way
- A **random forest** offers each split a random subset of `max_features` (default $\sqrt{n}$, rounded down), lowering ρ
- In code: `RandomForestClassifier()`, 100 trees by default

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Random forests, measured



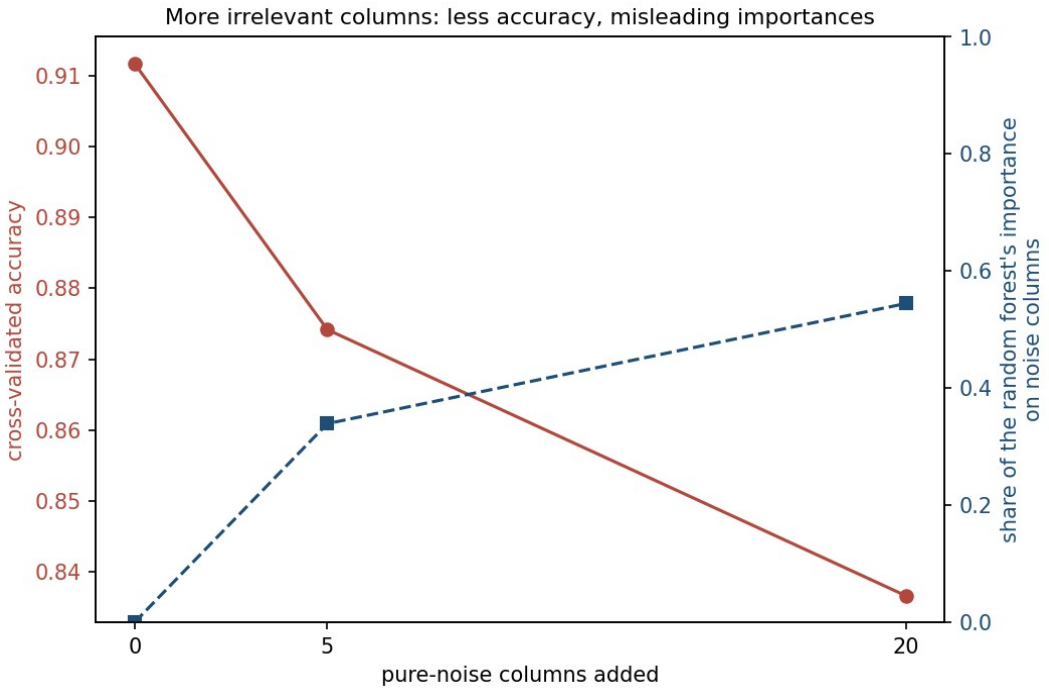
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Feature importance: what the forest claims to tell you

- **Feature importance**: the impurity reduction credited to each feature over every tree and split, scaled to add up to 1
- With only the two real features here, it names exactly those two
- What happens once we add columns of random numbers, unrelated to default?
- In code: `forest.feature_importances_`

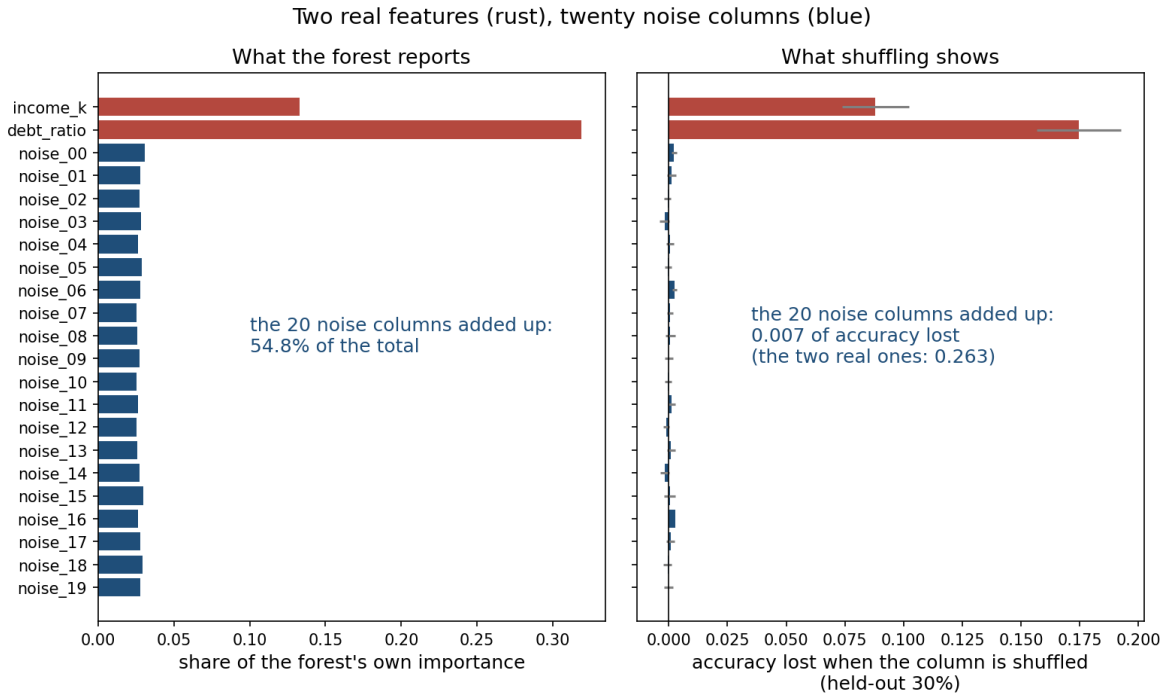
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

More columns, more noise



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Permutation importance



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Notebook 2, live

section, and its slides	you call (→ the number to come back with)
1. bagging from scratch, 27	`train_test_split`, your bagging against `BaggingClassifier`, `accuracy_score` → 0.872 and 0.872
2. why averaging works, 28-32	`oob_score=True`: `.oob_score_` against `cross_val_score` → 0.9117 both; 0.0026 over 12 seeds
3. variance, measured, 33-35	tree, `BaggingClassifier`, `RandomForestClassifier`, 30 splits → spread 0.0185, 0.0153, 0.0137
5. feature importance, 36-37	`.feature_importances_`, 0 to 20 noise columns → 54.5% on noise
6. checking the importances, 38	`permutation_importance`, test split → real 0.09 and 0.18, noise ≈ 0

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Gradient boosting: correcting mistakes in sequence

- Bagging and random forests build trees **independently**, then average
- **Boosting** builds them **in sequence**: each new tree corrects what the trees so far got wrong
- Each is a shallow **regression tree**, a **weak learner**: it predicts a number, a correction, not a class
- The learning rate α sets **how much** of each correction is applied

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Building the ensemble, one tree at a time, for squared error

$$F_0(x) = \bar{y}, \qquad F_t(x) = F_{t-1}(x) + \alpha \cdot h_t(x)$$

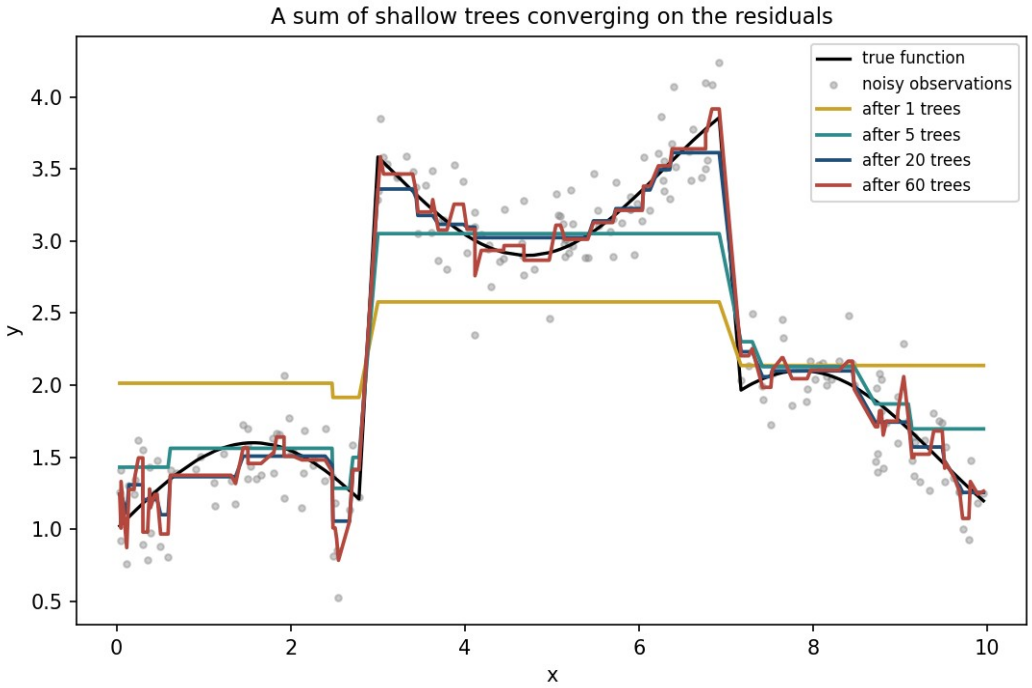
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What each tree is fit to

- For squared error, what remains unexplained is the **residual**, $y - F_{t-1}(x)$
- In general, **functional gradient descent**: h_t approximates the negative gradient of the loss with respect to F_{t-1}
- With $L = \frac{1}{2}(y - F)^2$, $-\partial L/\partial F = y - F$: the two coincide
- In code: `DecisionTreeRegressor(max_depth=2)`, fit to `y - prediction`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

A step plus a sine wave, fit by 60 shallow trees



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Reading the picture

- **1 tree**: mean squared error (MSE) against the true function 0.395, barely moves the flat guess
- **5 trees**: MSE 0.068, the coarse shape appears
- **20 trees**: MSE **0.012**, the ensemble’s best point
- **60 trees**: MSE rises back to 0.018, tracking noise, not signal

Classification: the same idea, a different gradient

- y is 0 or 1; the score F becomes a probability through lesson 4’s logistic function, $p = \sigma(F)$ (this σ is not the σ^2 of the variance slides)
- The loss is lesson 4’s **log-loss**
- Each tree fits the **pseudo-residual** y - p, the negative gradient: lesson 4’s error term
- Start: $p = 0.387$ for all, so the first tree fits $+0.613$ to defaulters and -0.387 to the rest

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

On the loan data

- `GradientBoostingClassifier`, defaults: **100 trees**, learning rate $\alpha = 0.1$
- Reaches **0.902 ± 0.020** cross-validated accuracy
- Already ahead of any single tree from earlier in the lesson

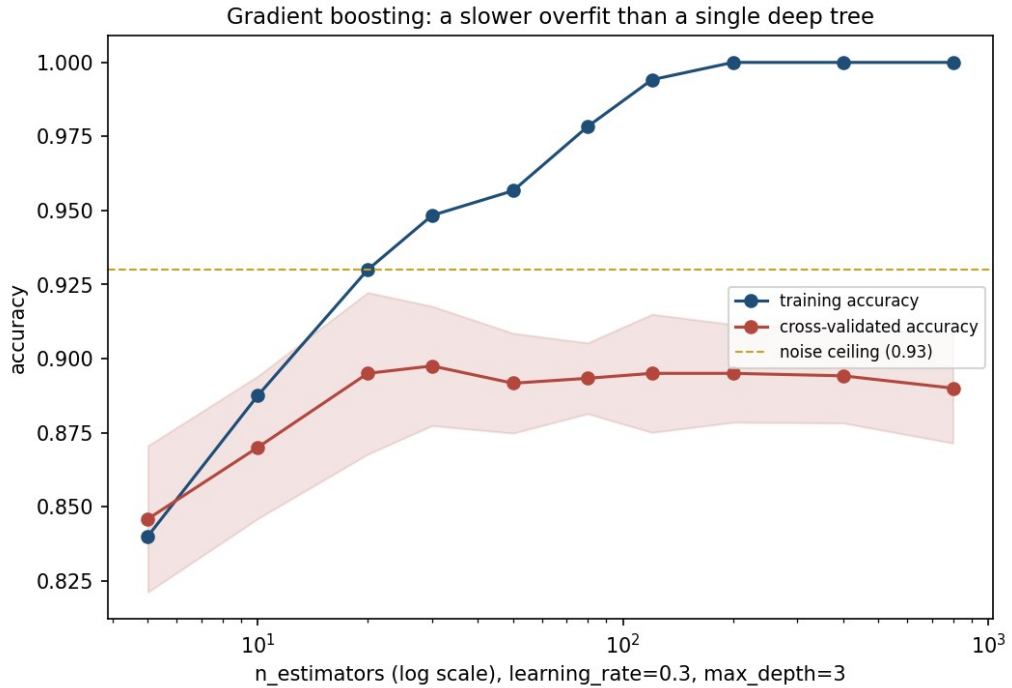
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Boosting never stops fitting on its own

`n_estimators` (α = 0.3, depth 3)	Training accuracy	Cross-validated accuracy
5	0.840	0.846 ± 0.025
10	0.888	0.870 ± 0.024
20	0.930	0.895 ± 0.027
30	0.948	**0.897 ± 0.020**
80	0.978	0.893 ± 0.012
200	1.000	0.895 ± 0.017
800	1.000	0.890 ± 0.019

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Training climbs to 1.000 and stays; the honest curve turns over first



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Fixing 120 trees, varying only the learning rate

Learning rate α	Training accuracy	Cross-validated accuracy
0.02	0.888	0.872 ± 0.025
0.05	0.914	0.897 ± 0.022
0.10	0.939	**0.902 ± 0.022**
0.30	0.994	0.895 ± 0.020
1.00	1.000	0.878 ± 0.020

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 3, live

section, and its slides	you call (→ the number to come back with)
1. boosting from scratch, 40-44	`DecisionTreeRegressor(max_depth=2)` on the residuals, 60 times → MSE 0.0119 at 20 trees
2. classification, 45-46	`GradientBoostingClassifier()` in `cross_val_score` → F0 = −0.461; 0.902
3. more trees, 47-48	the same, `learning_rate=0.3`, `n_estimators` from 5 to 800 → peak 0.897 at 30
4. learning rate, 49	the same with `n_estimators=120` and `learning_rate` from 0.02 to 1.0 → best 0.902 at 0.1
5. leaderboard, 51	every model in one `cross_val_score` loop → random forest 0.911

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Every method, on the same five folds

Model	Cross-validated accuracy
Majority baseline	0.613
Logistic regression	0.748
Single tree, unconstrained	0.852
Single tree, `max_depth = 8`	0.882
Gradient boosting, 30 trees	0.897
Bagging, 100 trees	0.904
Random forest, 100 trees	**0.911**
noise ceiling	*≈ 0.93*

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Three findings

- **On this dataset**, the ensembles land within 1.5 points of each other: **whether you ensemble** matters more than **which**
- The tuned single tree is close behind, 0.882 against 0.911, and the only model here a person can read
- The **unconstrained tree is the worst**, though the most flexible

How to choose, in practice

Situation	Reach for
The decision must be explained to the person it affects	A single tree, small enough to read; if even that is too opaque, none of these models
Tabular data, accuracy matters most, no explanation required	A random forest: the best score here, at its default settings
Accuracy matters, and there is time to tune the number of trees and the learning rate together	Gradient boosting, tuned on validation data
Many columns, few of them informative	Any tree method, but check its importances by permutation
Too few rows to set a validation set aside	A random forest, judged by its out-of-bag score

54% of the importance landed on noise

- 20 noise columns, 2 real ones: not small, not a bug
- **The menu**: 4 of 22 columns per split; in two splits of three (0.662), no real feature
- **Small groups**: with all 22 on offer, noise still takes **28%**, from deep splits
- The check: **permutation importance**, slide 38

What to take away

- A tree splits greedily on Gini impurity, matching scikit-learn
- Depth is the bias-variance dial: accuracy peaked at **depth 8**
- Bootstrap resampling leaves **36.8%** of rows out-of-bag: free validation
- **54% of a random forest’s importance landed on 20 noise columns**, today’s number

UNIVERSITÀ DEGLI STUDI DELL’AQUILA

Homework

- **Exercise 7**, discussed at the start of **Friday 27 November**
- `Exercises/07_trees_and_ensembles.md`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA